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UTILIZATION OF CARBON DIOXIDE EMISSIONS AS A FUEL SOURCE AND RELATED SYSTEMS AND METHODS

Abstract

Systems and methods of utilizing captured carbon dioxide emissions as a fuel source for the production of energy. The captured carbon dioxide emissions can be transformed into a carbon dioxide fluid having a higher energy state and stored as a fuel source, whereby the fuel source can be consumed at a desired time for the production of energy. The carbon dioxide fluid can be generated from a feedstock of the carbon dioxide emissions during periods of low energy demand, such as utilizing electricity during off-peak, which can then be used as a fuel source for energy production during periods of high energy demand. The electricity during off-peak preferably being from a renewable energy source, such as wind or solar power. The source of captured carbon dioxide emissions can be from an industrial source, such as the flue gas of a power plant, emissions from a cement plant, the emissions from a fermenter of an ethanol plant or brewery, or the combustion of fuels.

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Background/Summary

CLAIM FOR PRIORITY [0001] This is a National Stage Application which claims benefit to PCT International Application PCT/US2023/066440 filed May 1, 2023, which claims the benefit of priority of U.S. Provisional Patent Application No. 63/337,447 filed May 2, 2022, the subject matter of which is incorporated by reference in their entirety.

TECHNICAL FIELD

[0002] The present invention relates to systems and methods of utilizing carbon dioxide emissions from an industrial source as a fuel source, more particularly the carbon dioxide emissions transformed into a fuel source for the storage and production of energy.

BACKGROUND

[0003] Carbon dioxide (CO.sub.2) is a greenhouse gas whose concentration in the atmosphere has continued to grow since the Industrial Revolution. CO.sub.2 absorbs infrared radiation in the atmosphere, resulting in heat being trapped. Since CO.sub.2 traps heat in the atmosphere, an increased concentration in the atmosphere leads to an alteration of global climate by increasing global temperatures and altering weather patterns. These changes can cause widespread ecological damage, species extinction, flooding of low-lying coastal areas, and harm agriculture in certain areas.

[0004] Prior to the Industrial Revolution, global emissions of CO.sub.2 were very low. The growth in emissions of CO.sub.2 was still relatively slow until the mid-20th century with the world emitting 6 billion tons of CO.sub.2 in 1950. By 1990, this had almost quadrupled, reaching more than 22 billion tons. Since 1990, emissions of CO.sub.2 have continued to grow rapidly, with the global emissions reaching over 34 billion tons each year.

[0005] The combustion of hydrocarbon fossil fuels has been a mainstay in society for more than a century. According to the United States Environmental Protection Agency, hydrocarbon fossil fuels are the primary source of carbon dioxide (CO.sub.2) emissions, with global carbon emissions from hydrocarbon fossil fuels having increased significantly since 1900. Since 1970, CO.sub.2 emissions have increased by about 90%, with emissions from fossil fuel combustion and industrial processes contributing about 78% of the total greenhouse gas emissions increase from 1970 to 2011. In 2014, the top CO.sub.2 emitters were China, the United States, the European Union, India, the Russian Federation and Japan. Flue gas emissions-the emitted material produced when fossil fuels such as coal, oil, natural gas, or wood are burned for heat or power-contain pollutants, including CO.sub.2, nitrogen oxides and sulfur oxides.

[0006] With current environmental regulations, CO.sub.2 capture is very crucial for the survival of fossil fuel-fired power plants in the near future. Efforts to capture CO.sub.2 at some power plants have been successful, but the cost of installing and operating the required equipment is high. As such, very few power plants have carbon capture and storage (CSS) systems. In order for the sale of captured CO.sub.2 to become a profitable venture, the cost of capturing the CO.sub.2 from a flue gas must be reduced.

[0007] As the world moves towards clean energy initiative, carbon capture and utilization technologies are key to achieving net zero emissions. As CO.sub.2 emissions rise, CO.sub.2 capture and utilization technologies have been deemed necessary to reduce pollution and mitigate

the related climate effects. The goal of CO₂ capture technology is to provide a method of isolating CO₂ and reducing its emissions to the environment. The ideal long-term goal of such emissions reduction is to reach net negative emissions, where human activities balance out or are result in the net removal of CO₂ from the atmosphere. CO₂ utilization seeks to make this an economic and viable prospect by putting the CO₂ to work in stable and valuable tasks. Direct utilization uses the CO₂ as-is, without chemical conversion to other products. Widespread direct uses of CO₂ include use in food and beverages, fire extinguishers, concrete building materials, and CO₂ enhanced oil recovery. Indirect utilization uses the CO₂ as a feedstock in creating a more complex final product. Indirect utilization techniques primarily include the conversion of CO₂ to useful chemicals or fuels. The conversion of CO₂ to high energy density fuels is an attractive option to meeting the energy storage demands facing renewable energy.

[0008] The major challenge associated with utilizing CO₂ from waste streams is the cost of capturing it from those streams as opposed to acquiring CO₂ from natural sources. Large amounts of CO₂ can be obtained directly from natural gas reservoirs and industrial emissions, but in many cases the former has an economic advantage over the latter.

[0009] In addition to capturing CO₂ from fossil fuel-fired power plants, there has been a shift to renewable energy sources, such as wind, solar, thermal, hydro and biofuel. Since these renewable energy sources draw power from natural sources, there are problems associated with renewable energy sources, including intermittency and predictability problems associated with natural cycles that can limit the use or efficiencies of renewable energies. For example, energy production from solar and wind may have unpredictable lulls that define the intermittency of renewable energy sources. The intermittency problem ultimately manifests economically as either curtailment, large supply with insufficient demand, or elevated price, large demand with insufficient supply. This intermittency is contrasted by the constant power output that can be generated by hydrocarbon fossil fuel-based power plants. Accordingly, intermittency and predictability problems of renewable energy sources limits the growth of renewable energy.

[0010] Despite these drawbacks, renewable energy sources have seen continued growth. For example, ethanol is a major hydrogen-rich liquid transport fuel around the world that has seen continued growth since the 1990's. Worldwide consumption of ethanol in 2005 was an estimated 12.2 billion gallons. The U.S. Department of Energy reported a continued growth in global ethanol production, with global production bypassing 20 billion gallons each year since 2009 and reaching 29.03 billion gallons in 2019. The majority of ethanol fuel is produced via traditional yeast-based fermentation processes that use crop derived carbohydrates, such as sucrose extracted from sugarcane or starch extracted from grain crops, as the main carbon source. However, the production of CO₂ during fermentation represents an opportunity to utilize the CO₂ from ethanol fuel production, which has the potential to contribute to reducing greenhouse gas emissions from a currently balanced state where the current CO₂ emissions are taken up by the following year's crop to a carbon negative state. A typical ethanol plant producing 50 million gallons of ethanol per year will also produce roughly 150,000 metric tons of CO₂ per year.

[0011] Another problem associated with energy production relates to the unpredictability and fragility of the supply chain of energy sources globally, as evidenced by the COVID-19 pandemic and recent Russian-Ukraine conflict that has affected global supplies, including energy production sources. Still further, the U.S. power grid has continued to experience increased problems with the number of power outages rising annually. While there are a lot of factors contributing to the rise of power grid problems across the country, the chronic congestion on the long-distance transmission lines is a major contributor. Without a consistent, fail-proof source of power that can be provided at a local level, many businesses face damaging downtime when mission-critical functions cannot continue.

[0012] Accordingly, there is a need to address the release of CO₂ emissions into the

atmosphere to avoid a runaway greenhouse effect from the use of fossil fuels and/or renewable fuels in the production of power and energy. There is also a need for the economical capture and storage of CO₂ emissions. There is also a need to utilize captured CO₂ emissions in a manner that minimizes or produces no net greenhouse gas emissions. There is further a need to utilize captured CO₂ emissions in a manner that reduces costs, or alternatively, provides a returnable profit from the use of CO₂ as a traditional emitted waste product. Still even further, with the unpredictability and fragility of the supply chain of energy sources globally and the vulnerability of the power grid, there is also a need for an environmentally friendly on-demand energy source that can be provided locally, efficiently and in a cost-effective manner.

SUMMARY

[0013] The present disclosure is directed at systems and methods of addressing carbon dioxide emissions while simultaneously addressing the intermittency issue of renewable energy sources by utilizing carbon dioxide emissions as a feedstock that is transformed into a carbon dioxide fluid having a higher energy state that can be stored as a fuel source, which can be stored, whereby the fuel source can be consumed at a desired time for the production of energy.

[0014] In some aspects, carbon dioxide emissions are captured as a feedstock and transformed into a carbon dioxide fluid having a higher energy state and stored as a fuel source during periods of low energy demand (off-peak), which can then be used as a fuel source for energy production during periods of high energy demand (peak load).

[0015] In some aspects, excess electricity during off-peak can be used to transform a captured carbon dioxide emission feedstock stream into the higher state carbon dioxide fluid. The higher state carbon dioxide fluid can be stored in a storage reservoir as a fuel source having stored potential energy until there is a need for energy production. In the storage reservoir, the stored potential energy of the carbon dioxide fuel source is capable of being substantially maintained for a period of time. When there is a need for energy production, the stored potential energy of the carbon dioxide fuel source can be utilized for the production of energy by using the carbon dioxide fuel source as an input fuel source in a device for energy production, such as by spinning a turbine.

[0016] The present disclosure is directed at systems and methods of addressing gaseous carbon dioxide emissions by capturing gaseous carbon dioxide, transforming the gaseous carbon dioxide into a physical form having a higher stored potential energy, storing the captured carbon dioxide in the form having higher stored potential energy as a fuel source, and utilizing the potential energy of the stored carbon dioxide fuel source as an input fuel source for the production of energy. In some aspects, the higher stored potential energy of the carbon dioxide fuel source can be consumptively harnessed into mechanical work for the production of energy. In some aspects, the physical form of carbon dioxide having higher stored potential energy is a carbon dioxide fluid. In some aspects, the carbon dioxide fuel source is transformed to a physical form having a lower stored potential energy than the carbon dioxide liquid. In some aspects, the physical form of the carbon dioxide having the lower stored potential energy is gaseous carbon dioxide, solid carbon dioxide, or a combination thereof.

[0017] In some aspects, the industrial source of carbon dioxide is carbon dioxide emissions from the flue gas of a power plant, such as during the use of hydrocarbon fossil fuels and/or renewable fuels, such as renewable biofuels.

[0018] In some aspects, the industrial source of carbon dioxide is carbon dioxide emissions, which are generated in a fermenter during the production of ethanol in an ethanol plant, such as carbon dioxide emissions released during the fermentation process. In some other aspects, the industrial source is carbon dioxide generated in a fermenter of a brewery, such as carbon dioxide emissions released during the fermentation process.

[0019] In some aspects, a system for the capture and utilization of carbon dioxide as a fuel source from a biomass feedstock for the production of energy comprises a feedstock stream comprising captured carbon dioxide emissions having a first energy state, a transformation means for

transforming the captured carbon dioxide emissions of the feedstock stream into a carbon dioxide fluid having a second energy state that is higher than the first energy state of the captured carbon dioxide emissions of the feedstock stream, a storage means for storing the carbon dioxide fluid as a fuel source, a consumptive means that is capable of using the fuel source to produce an output energy and a discharged carbon dioxide, the discharged carbon dioxide having a different physical form than the carbon dioxide of the fuel source, and an optional means for recapturing the discharged carbon dioxide.

[0020] In some aspects, a system for the capture and utilization of carbon dioxide emissions as a fuel source for the production of energy comprises a feedstock stream comprising captured carbon dioxide emissions, wherein the captured carbon dioxide emissions are provided from fermentation of a biomass feedstock in a fermenter of an ethanol plant, a first device that is capable of transforming the captured carbon dioxide emissions of the feedstock stream into a carbon dioxide fluid, such as a compressor, wherein the carbon dioxide fluid has a higher energy state than the captured carbon dioxide emissions, a vessel for receiving and storing the carbon dioxide fluid as a fuel source, a second device that is capable of using the fuel source to produce an output energy and a discharged carbon dioxide, such as a turbine, the discharged carbon dioxide having a different physical form than the carbon dioxide fuel source, and an optional third device that is capable of recapturing the discharged carbon dioxide.

[0021] In some aspects, the fuel source is stored below its critical temperature and above its critical pressure, such that the fuel source is stored in the form of a liquid. In some preferred aspects, the fuel source is stored at a temperature between -56.6°C . and 31°C . and above a pressure of 5.2 bar. In some preferred aspects, the fuel source is stored at a pressure between about 5.2 bar and about 100 bar, in some aspects between about 10 bar and about 90 bar, in some aspects between about 25 bar and about 85 bar, in some aspects between about 40 bar and about 80 bar, and in some preferable aspects between about 45 bar and about 65 bar. In some preferred aspects, the fuel source is stored at a temperature between about -30°C . and about 30°C ., in some aspects between about -25°C . and about 25°C ., in some aspects between about -20°C . and about 20°C ., and in some preferred aspects between about -18°C . and about 18°C .

[0022] In some aspects, fuel source is stored above its critical temperature and above its critical pressure, such that at least a portion of fuel source is stored in the form of a supercritical liquid. In some preferred aspects, fuel source is stored above a temperature of 31°C . and above a pressure of 73.8 bar. In some preferred aspects, fuel source is stored at a pressure between about 73.8 bar and about 1,000 bar, more preferably between about 73.8 bar and about 500 bar, more preferably between about 73.8 bar and about 250 bar, and even more preferably between about 73.8 bar and about 125 bar.

[0023] In some aspects, a source of at least a portion of the feedstock stream comprising captured carbon dioxide emissions is provided from a fermenter.

[0024] In some aspects, the fermenter is from an ethanol plant or a brewery.

[0025] In some aspects, the feedstock stream comprising captured carbon dioxide emissions from the fermenter is produced as a biogenic waste stream from a biomass feedstock during fermentation of the biomass feedstock at the ethanol plant.

[0026] In some aspects, the biomass feedstock is selected from the group consisting of corn, sugar cane, bagasse, miscanthus, sugar beet, sorghum, grain, switchgrass, barley, hemp, kenaf, potatoes, sweet potatoes, cassava, sunflower, fruit, molasses, stover, wheat, straw, cotton and algae.

[0027] In some aspects, the biomass feedstock comprises corn kernels.

[0028] In some aspects, the transformation means comprises a compressor.

[0029] In some aspects, the transformation means utilizes an energy input for transforming the captured carbon dioxide emissions of the feedstock stream into the carbon dioxide fluid.

[0030] In some aspects, at least a portion of the energy input is a renewable energy input.

[0031] In some aspects, the renewable energy input is provided from wind power, solar power, the

output energy, or a combination thereof.

[0032] In some aspects, at least a portion of the carbon dioxide fluid comprises liquid carbon dioxide.

[0033] In some aspects, the consumptive means comprises a turbine, a turboexpander, or a combination thereof.

[0034] In some aspects, at least a portion of the discharged carbon dioxide comprises gaseous carbon dioxide, solid carbon dioxide, or a combination thereof.

[0035] In some aspects, the output energy is in the form of electricity.

[0036] In some aspects, the electricity is capable of being fed into an electrical power grid.

[0037] In some aspects, the output energy is produced by utilizing the higher second energy state of the carbon dioxide fuel source in a turbine to produce electricity and the discharged carbon dioxide having a third energy state that is lower than the second energy state of the carbon dioxide fuel source.

[0038] In some aspects, the transformation means generates a source of heat during the process of transforming the captured carbon dioxide emissions of the feedstock stream into the carbon dioxide fluid.

[0039] In some aspects, the system further comprises at least one heat exchanger, wherein the at least one heat exchanger is capable of utilizing the source of heat generated.

[0040] In some aspects, the least one heat exchanger utilizes the source of heat in an ethanol plant.

[0041] In some aspects, a system for the capture and utilization of carbon dioxide emissions as a fuel source for the production of energy, the system comprise a feedstock stream comprising captured carbon dioxide emissions, a compressor that is capable of transforming the captured carbon dioxide emissions of the feedstock stream into a carbon dioxide fluid, wherein the carbon dioxide fluid has a higher energy state than the captured carbon dioxide emissions, a vessel for receiving and storing the carbon dioxide fluid as a fuel source, a device capable of using the fuel source to produce an output energy and a discharged carbon dioxide, the discharged carbon dioxide having a different physical form than the carbon dioxide fuel, and an optional means for recapturing the discharged carbon dioxide.

[0042] In some aspects, a method of capturing and utilizing captured carbon dioxide emissions as a fuel source for the production of energy, the method comprises capturing carbon dioxide emissions from a source to provide a feedstock stream comprising captured carbon dioxide emissions, feeding the feedstock stream to a compressor having an input energy source, transforming the captured carbon dioxide emissions of the feedstock stream into a carbon dioxide fluid having stored potential energy, the carbon dioxide fluid having a different physical form than the carbon dioxide gas, storing the carbon dioxide fluid as a fuel source in a usable state to maintain the stored potential energy of the carbon dioxide fuel source, producing an output energy and a discharged carbon dioxide from the stored potential energy of the carbon dioxide fuel source, the discharged carbon dioxide having a different physical form than the carbon dioxide fuel source, and optionally recapturing the discharged carbon dioxide.

[0043] The above summary is not intended to describe each illustrated embodiment or every implementation of the subject matter hereof. The figures and the detailed description that follow more particularly exemplify various embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0044] Subject matter hereof may be more completely understood in consideration of the following detailed description of various embodiments in connection with the accompanying figures, in which:

[0045] FIG. **1** is a schematic diagram illustrating a system capable of utilizing captured CO.sub.2 as a fuel source for the production of energy, wherein the system capable of conducting a method of capturing carbon dioxide emissions to provide a source of captured carbon dioxide, converting the captured carbon dioxide into a different physical form having a high potential energy, the different physical form of carbon dioxide configured to be capable of being stored, and utilizing the different form of carbon dioxide as a fuel source for the generation of energy, according to certain embodiments of the present invention.

[0046] FIG. **2** is a schematic diagram illustrating a system capable of utilizing captured CO.sub.2 as a fuel source for the production of energy, wherein the system capable of conducting a method of capturing carbon dioxide emissions from an ethanol plant to provide a feedstock of captured carbon dioxide, converting the feedstock of captured carbon dioxide into a different physical form having a high potential energy, the different physical form of carbon dioxide configured to be capable of being stored, and utilizing the different form of carbon dioxide as a fuel source for the generation of energy, according to certain embodiments of the present invention.

[0047] FIG. **3** is a schematic diagram illustrating a system capable of utilizing captured CO.sub.2 as a fuel source for the production of energy, wherein the system capable of conducting a method of capturing carbon dioxide emissions from an ethanol plant to provide a source of captured carbon dioxide, converting the captured carbon dioxide into a different physical form having a high potential energy, the different physical form of carbon dioxide configured to be capable of being stored, and utilizing the different form of carbon dioxide as a fuel source for the generation of energy, wherein additional energy can be added to the stored carbon dioxide prior to the stored potential energy being harnessed into mechanical work for the production of energy, according to certain embodiments of the present invention.

[0048] While various embodiments are amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings and will be described in detail. It should be understood, however, that the intention is not to limit the claimed inventions to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the subject matter as defined by the claims.

DETAILED DESCRIPTION OF THE DRAWINGS

[0049] The present disclosure is directed at systems and methods of utilizing CO.sub.2 emissions as a fuel source by transforming the captured CO.sub.2 emissions into a physical form having a higher energy state than the CO.sub.2 emissions, and the higher state CO.sub.2 can be stored as a fuel source until the stored potential energy of the fuel source is consumptively utilized at a desired time for the production of energy.

[0050] Referring generally to FIGS. **1-3**, a system **100** for the utilization of captured CO.sub.2 as a fuel source for the production of energy is illustrated. According to certain embodiments, the system **100** comprises a source of captured CO.sub.2 **110**, which can be provided as a feedstock stream **113** of the captured CO.sub.2 to a means for transforming the feedstock stream **113** of the captured CO.sub.2 into a physical form having a higher energy state. In some preferred aspects, the source of captured CO.sub.2 **110** comprises captured CO.sub.2 emissions. In some other aspects, the source of captured CO.sub.2 **110** comprises CO.sub.2 captured from the atmosphere. In some preferred aspects, the higher energy state comprises CO.sub.2 fluid **123**.

[0051] In some preferred aspects, the CO.sub.2 fluid **123** comprises liquid CO.sub.2. In some other aspects, the CO.sub.2 fluid **123** comprises supercritical CO.sub.2. In some other aspects the CO.sub.2 fluid **123** comprises liquid CO.sub.2, supercritical CO.sub.2, gaseous CO.sub.2, or a combination thereof.

[0052] The CO.sub.2 fluid **123** preferably has a higher energy state than the feedstock stream **113** of captured CO.sub.2 prior to the transformation, such that the source of captured CO.sub.2 **110** has a lower energy state than the CO.sub.2 fluid **123**. In some preferred aspects, the feedstock stream

113 of captured CO.sub.2 prior to transformation to the CO.sub.2 fluid **123** has approximately an ambient temperature and/or an ambient pressure. The means for transforming the feedstock stream **113** of the captured CO.sub.2 into the CO.sub.2 fluid **123** preferably increases the fluid pressure of the feedstock stream **113** of the captured CO.sub.2 to provide the CO.sub.2 fluid **123** having the higher energy state. In some preferred aspects, the fluid pressure of the feedstock stream **113** of the captured CO.sub.2 is increased to at least about 6.5 MPa (65 bar/950 psi) in order to provide the CO.sub.2 fluid **123** having the higher energy state.

[0053] In some aspects, the means for transforming the feedstock stream **113** of captured CO.sub.2 into the CO.sub.2 fluid **123** having the higher energy state comprises compressor **120**, which increases the fluid pressure of the feedstock stream **113** of captured CO.sub.2. In some preferred aspects, the means for transforming the feedstock stream **113** of CO.sub.2 into the CO.sub.2 fluid **123** having the higher energy state, such as compressor **120**, increases the fluid pressure of the feedstock stream **113** of captured CO.sub.2 to at least about 5.2 bar, in some aspects at least about 10 bar, in some aspects at least about 25 bar, in some aspects at least about 40 bar, in some aspects at least about 50 bar, and in some aspects at least about 55 bar, in order to provide the CO.sub.2 fluid **123** having the higher energy state.

[0054] In some preferred aspects, the means for transforming the feedstock stream **113** of CO.sub.2 into the CO.sub.2 fluid **123** having the higher energy state, such as compressor **120**, increases the fluid pressure of the feedstock stream **113** of captured CO.sub.2 to between about 5.2 bar and about 100 bar, in some aspects between about 10 bar and about 90 bar, in some aspects between about 25 bar and about 85 bar, in some aspects between about 40 bar and about 80 bar, and in some preferable aspects between about 45 bar and about 65 bar, in order to provide the CO.sub.2 fluid **123** having the higher energy state.

[0055] In some aspects, compressor **120** may be a multistage compressor, piston-based, turbine or gravity well. Compressor **120** can be used to produce the CO.sub.2 fluid **123** having the higher state by compressing the feedstock stream **113** of the captured CO.sub.2 to pressures between about 5.2 bar and about 100 bar, in some aspects between about 10 bar and about 90 bar, in some aspects between about 25 bar and about 85 bar, in some aspects between about 40 bar and about 80 bar, and in some preferable aspects between about 45 bar and about 65 bar. The resulting pressurized fluid will also have an increased temperature above ambient temperature, typically above the critical point of the pressurized fluid, in some aspects above 100° C., in some aspects above 250° C., in some aspects above 400° C. and in some aspects up to about 510° C. The resulting pressurized fluid is preferably cooled to a temperature below the critical point, which in some aspects is about 31° C., in some aspects between about 10° C. and about 30° C., preferably between about 20° C. and about 28° C.

[0056] In some preferred aspects, the feedstock stream **110** of the captured CO.sub.2 prior to transformation comprises gaseous CO.sub.2, which is transformed into the higher energy state. The feedstock stream **110** of the captured CO.sub.2 prior to transformation preferably comprises at least 70% gaseous CO.sub.2, in some aspects at least 75% gaseous CO.sub.2, in some aspects at least 80% gaseous CO.sub.2, in some aspects at least 85% gaseous CO.sub.2, in some aspects at least 90% gaseous CO.sub.2, in some aspects at least 95% gaseous CO.sub.2, in some aspects at least 98% gaseous CO.sub.2, in some aspects at least 99% gaseous CO.sub.2, and in some aspects at least 99.9% gaseous CO.sub.2. The feedstock stream **110** of the captured CO.sub.2 provided to the means for transforming the captured CO.sub.2 into a physical form having a higher energy state can further comprise other components, such as water vapor.

[0057] In some aspects, feedstock stream **110** of the captured CO.sub.2 prior to transformation has approximately ambient temperature and approximately ambient pressure. In some preferred aspects, feedstock stream **110** of the captured CO.sub.2 prior to transformation has a temperature between about 5° C. and about 30° C., in some aspects between about 10° C. and about 28° C., and in some preferred aspects between about 15° C. and about 25° C. In some preferred aspects,

feedstock stream **110** of the captured CO.sub.2 prior to transformation has a pressure between about 1bar and about 1.5 bar.

[0058] Gaseous CO.sub.2 can be liquified under pressure provided its temperature is below 31° C., which is referred to as the critical point. If compressed and cooled below the critical point, a colorless fluid having approximately the same density of water is produced. CO.sub.2 will remain in the liquid form as long as its temperature remains below the critical point, but will return to the gaseous state if its temperature rises above the critical point, regardless of the pressure applied.

[0059] In some preferred aspects, CO.sub.2 fluid **123** provided from the transformation process comprises liquid CO.sub.2. The CO.sub.2 fluid **123** preferably comprises at least 70% liquid CO.sub.2, in some aspects at least 75% liquid CO.sub.2, in some aspects at least 80% liquid CO.sub.2, in some aspects at least 85% liquid CO.sub.2, in some aspects at least 90% liquid CO.sub.2, in some aspects at least 95% liquid CO.sub.2, in some aspects at least **98%** liquid CO.sub.2, in some aspects at least 99% liquid CO.sub.2, and in some aspects at least 99.9% liquid CO.sub.2. A minority percentage of the CO.sub.2 fluid **123** having the higher energy state can be provided in in another phase, such as supercritical CO.sub.2, gaseous CO.sub.2, or a combination thereof.

[0060] In some alternative aspects, the CO.sub.2 fluid **123** provided from the transformation process comprises supercritical CO.sub.2. The CO.sub.2 fluid **123** in this alternative embodiment comprises at least 70% supercritical CO.sub.2, in some aspects at least 75% supercritical CO.sub.2, in some aspects at least 80% supercritical CO.sub.2, in some aspects at least 85% supercritical CO.sub.2, in some aspects at least 90% supercritical CO.sub.2, in some aspects at least 95% supercritical CO.sub.2, in some aspects at least 98% supercritical CO.sub.2, in some aspects at least 99% supercritical CO.sub.2, and in some aspects at least 99.9% supercritical CO.sub.2. A minority percentage of the CO.sub.2 fluid **123** having the higher energy state can be provided in in another phase, such as liquid CO.sub.2, gaseous CO.sub.2, or a combination thereof.

[0061] The means for transforming the feedstock stream **113** of CO.sub.2 into the CO.sub.2 fluid **123** having the higher energy state, such compressor **120**, can be powered by an energy input **130**. In some aspects, energy input **130** is electricity from the electrical grid, methane gas combustion, or sourced from renewable energy, such as wind, solar, biofuels and the like. In some aspects, the feedstock stream **113** of captured CO.sub.2 is transformed into fluid CO.sub.2 **123** having a higher energy state during periods of low energy demand (off-peak). In some aspects, a renewable energy source is utilized to transform the feedstock stream **113** of CO.sub.2 to the fluid CO.sub.2 **123** having a higher energy state during off-peak when the demand for energy is less than the supply. In some aspects, the use of off-peak energy from a renewable energy source, such as wind or solar, enables the use of the renewable energy source that would otherwise be lost or not otherwise fully utilized. Not only does the use of off-peak energy avoid the higher charges of peak demand, but using energy during off-peak can also avoid grid consumption during peak demand. This can provide grid security and stability improvements for on-demand energy storage and power backup, including black start capability, load balancing, determent, and other advantages.

[0062] The CO.sub.2 fluid **123** having the higher state energy is a fuel source **145**. In some preferred aspects, CO.sub.2 fluid **123** is a fuel source **145** capable of being stored for use during a desired time. In some aspects, fuel source **145** can be stored in a storage reservoir **140**.

[0063] In some aspects, storage reservoir **140** can be a specially constructed tank that can maintain the CO.sub.2 fluid **123** under pressure. In some preferred aspects, the storage reservoir **140** is a specially constructed steel tank, more preferably a steel tank lined with a specialty alloy. In some preferred aspects, storage reservoir **140** is temperature controlled to maintain the CO.sub.2 fluid **123** below the critical point. In some preferred aspects, storage reservoir **140** is adiabatically insulated, such as in a temperature-controlled structure, insulated or the like. In some aspects, storage reservoir **140** comprises a tank having a volume between about 1,000 and about 100,000 gallons, preferably between about 2,500 and about 75,000 gallons, preferably between about 5,000

and about 50,000 gallons, and in some preferred aspects between about 7,500 and about 15,000 gallons. In some aspects, storage reservoir **140** for containing and storing fuel source **145** can be transported from one location to another location. In some other aspects, storage reservoir **140** for containing and storing fuel source **145** can be provided at an elevated height, such that the potential energy of the CO.sub.2 fluid **123** is increased. In some other aspects, storage reservoir **140** for containing and storing fuel source **145** can be a steel tank that is at least partially submerged within the surrounding ground.

[0064] In some aspects, storage reservoir **140** for containing and storing fuel source **145** can be a specially constructed storage tank maintained at ambient temperature below the critical point.

[0065] In some aspects, fuel source **145** is stored below its critical temperature and above its critical pressure, such that at least a portion of fuel source **145** is stored in the form of a liquid. In some preferred aspects, fuel source **145** is stored at a temperature between -56.6°C . and 31°C . and above a pressure of 5.2 bar. In some preferred aspects, fuel source **145** is stored at a pressure between about 5.2 bar and about 100 bar, in some aspects between about 10 bar and about 90 bar, in some aspects between about 25 bar and about 85 bar, in some aspects between about 40 bar and about **80** bar, and in some preferable aspects between about 45 bar and about 65 bar. In some preferred aspects, fuel source **145** is stored at a temperature between about -30°C . and about 30°C ., in some aspects between about -25°C . and about 25°C ., in some aspects between about -20°C . and about 20°C ., and in some preferred aspects between about -18°C . and about 18°C .

[0066] In some aspects, fuel source **145** is stored above its critical temperature and above its critical pressure, such that at least a portion of fuel source **145** is stored in the form of a supercritical liquid. In some preferred aspects, fuel source **145** is stored above a temperature of 31°C . and above a pressure of 73.8 bar. In some preferred aspects, fuel source **145** is stored at a pressure between about 73.8 bar and about 1,000 bar, more preferably between about 73.8 bar and about 500 bar, more preferably between about 73.8 bar and about 250 bar, and even more preferably between about 73.8 bar and about 125 bar.

[0067] In some preferred aspects, storage reservoir **140** stores fuel source **145** at a pressure between about 45 and about 65 bar and a temperature between about 15°C . and 31°C ., more preferably between about 18°C . and 31°C ., more preferably between about 21°C . and about 26°C . In some more preferred aspects, storage reservoir **140** stores fuel source **145** at a pressure between about 60 and about 65 bar and a temperature maintained between about 18°C . and about 25°C . In some other aspects, storage reservoir **140** is a specially constructed storage tank that is heavily insulated and equipped with refrigeration units to hold the internal tank temperature between -35°C . and -15°C . and pressures between 12 to 25 bar.

[0068] Storage reservoir **145** containing fuel source **145** can also be transported from a first location to a second location, such as in insulated road tankers or trailers, fuel source **145** being simply transferred from a mobile tank to a static tank, or vice versa, by pumping or gravity feed. Fuel source **145** may also be transported from a first location to a second location by a pipeline. In some aspects, the first location is the means for transforming the feedstock stream **113** of CO.sub.2 into the CO.sub.2 fluid **123** having the higher energy state, such compressor **120**. In some other aspects, the first location is storage reservoir **140**. In some aspects, storage reservoir **140** may comprise a pipeline, and in some other storage reservoir **145** may consist essentially of a pipeline, particularly where the distance between the proximate location of the transformation and subsequent utilization of fuel source **145** is greater than 0.5 mile, in some aspects at least 0.5 mile to about 25 miles, in some aspects at least 0.5 mile to about 20 miles, in some aspects at least **0.5** mile to about 15 miles, in some aspects at least 0.5 mile to about 12 miles, in some aspects at least 0.5 mile to about 10 miles, in some aspects at least 0.5 mile to about 5 miles, in some aspects at least 1 mile to about 25 miles, in some aspects at least 1 mile to about 20 miles, in some aspects at least 1 mile to about 15 miles, in some aspects at least 1 mile to about 12 miles, in some aspects at least 1 mile to about 10 miles, and in some

aspects at least 1 mile to about 5 miles.

[0069] In some preferred aspects, the pipeline comprises high-strength steel pipe, preferably a carbon steel material engineered to meet standards set by the U.S. Department of Transportation's Pipeline and Hazardous Materials Safety Administration (PHMSA) relating to pipeline transportation of carbon dioxide. The pipeline can have a diameter between about 0.5 inches to about 48 inches, preferably between 4 inches and 24 inches, with a mainline pipeline between the first location and the second location preferably between 16 inches and 48 inches in diameter, in some aspects preferably between 4 inches and 24 inches, and lateral pipelines delivering fuel source 145 from the mainline preferably between 6 inches and 16 inches in diameter, in some aspects preferably between 4 inches and 12 inches.

[0070] When stored in storage reservoir 140, the CO.sub.2 fluid 123 is a fuel source 145, as the stored CO.sub.2 liquid 123 has stored potential energy that can produce an output energy to do mechanical work when converted to a different CO.sub.2 state. In some preferred aspects, the stored CO.sub.2 fuel source 145 can be fed into a consumptive means or device 150 that is capable of using the stored potential energy of the fuel source to produce an output energy 160 and a discharged CO.sub.2 170. In some preferred aspects, stored CO.sub.2 fuel source 145 can be fed to device 150 that is capable of using the stored potential energy of CO.sub.2 fuel source 145 to produce an output energy 160 by doing mechanical work and also provide discharged CO.sub.2 170. Since fuel source 145 has a higher energy density than gaseous CO.sub.2, fuel source 145 undergoing gas expansion when being fed into device 150 does mechanical work and produces the discharged CO.sub.2 170. Discharged CO.sub.2 170 has a different energy state that is lower than the higher energy state of fuel source 145, such that discharged CO.sub.2 170 has a lower potential energy level than fuel source 145. In some preferred aspects, device 150 is a turbine that has an output shaft connected to a generator or an output to drive a compressor, each of which produce energy. In some preferred aspects, stored potential energy of CO.sub.2 fuel source 145 produces an output energy 160 by spinning the turbine, such that output energy 160 is electricity generated by the generator connected to the turbine, which use of the higher energy state of the stored CO.sub.2 fuel source 145 also provides discharged CO.sub.2 170 having the lower energy state. In some preferred aspects, the output energy 160 of device 150 is in the form of electricity. In some other aspects, the output energy 160 in the form of electricity can be fed into a power grid 180, such as an electrical grid 180.

[0071] In some aspects, device 150 comprises a turboexpander. Fuel source 145 fed into the turboexpander reduces gas pressure and converts the resulting kinetic energy into energy production, particularly electrical energy. In particular, work is extracted from the expansion of the fuel source 145 in the turboexpander via a turbine. The kinetic energy (work) produced by the turbine can be used to drive a compressor or generator to produce energy, and the exhausting flow of discharged CO.sub.2 170 has a lower pressure and substantially lower temperature than CO.sub.2 fuel source 145 fed into the turbo-expander.

[0072] In some particular aspects, as illustrated in FIG. 3, CO.sub.2 fuel source 145 is heated to an increased temperature, such as temperatures between about 400° C. and about 450° C., prior to being fed into device 150, whereby the increased temperature prior to being fed to device 150 increases output energy production 160.

[0073] In some preferred, as illustrated in FIG. 3, CO.sub.2 fuel source 145 is heated to an increased temperature, such that the temperature of CO.sub.2 fuel source 145 at an inlet of device 150, such as at an inlet of a turbine or turboexpander, is between about 400° C. and about 450° C. and a pressure between about 40 bar and about 80 bar, in some aspects between about 50 bar and about 75 bar, and in some preferable aspects between about 55 bar and about 70 bar, whereby CO.sub.2 fuel source 145 expands at the point of device 150 to facilitate the generation of electrical power by increasing output energy production 160.

[0074] In some aspects, the stored CO.sub.2 fluid 123 is utilized as a fuel source 145 for energy

production during periods of high energy demand (peak load).

[0075] In some aspects, the discharged CO.sub.2 **170** is in a gaseous state, solid state, or a combination thereof. For instance, if the pressure of CO.sub.2 fluid **123** is released, at least a portion of that CO.sub.2 fluid **123** will change to the solid state and the remainder will revert to the gaseous state. In some aspects, at least a portion of the discharged CO.sub.2 **170** may be recaptured as solid CO.sub.2 having a temperature at or below the triple point temperature of -56.57°C . and a pressure at or below the triple point pressure of 5.18 bars. In some aspects, at least a portion of the discharged CO.sub.2 **170** comprises gaseous CO.sub.2, such that the pressure is less than the CO.sub.2 fluid **123** at the same relative temperature.

[0076] In some aspects, the recaptured solid-state CO.sub.2 of discharged CO.sub.2 **170** can be formed into dry ice blocks or pellets. In some alternative aspects, the recaptured solid-state CO.sub.2 of discharged CO.sub.2 **170** may be allowed to revert to the gaseous state.

[0077] In some aspects, the gaseous CO.sub.2, or a portion thereof, of CO.sub.2 fluid **123** used as fuel source **145** and providing discharged CO.sub.2 **170** can be recaptured. The recaptured CO.sub.2 gas can provide a portion of the feedstock stream **113** for the transformation into fuel source **145**, such that the utilization of the fuel source **145** can be repeated. In the instance of gaseous CO.sub.2 of discharged CO.sub.2 **170** being recaptured and provided as a portion of feedstock stream **113**, system **100** is always releasing a comparable amount of CO.sub.2, such that system **100** is an open system and not a closed loop system.

[0078] In some preferred aspects, the gaseous state CO.sub.2 and/or solid-state CO.sub.2 of discharged CO.sub.2 **170** can be recaptured for commercial use **190**. For instance, the gaseous state CO.sub.2 will have undergone a purification process during the transformation and utilization process, such that the gaseous state CO.sub.2 can be recaptured and used as industrial grade CO.sub.2 having a purity of at least 99.5%, in some aspects medical grade CO.sub.2 having a purity of at least 99.5%, in some aspects bone dry grade CO.sub.2 having a purity of at least 99.8%, in some aspects food grade CO.sub.2 having a purity of at least 99.9%, in some aspects beverage grade CO.sub.2 having a purity of at least 99.9%, in some aspects anaerobic grade CO.sub.2 having a purity of at least 99.95%, and in some aspects research grade CO.sub.2 having a purity of at least 99.999%.

[0079] In some alternative aspects, the solid state and/or gaseous state CO.sub.2 of discharged CO.sub.2 **170** can be recaptured and used for carbon sequestration.

[0080] In some aspects, the source of captured CO.sub.2 **110** for system **100** can be an industrial source of CO.sub.2, such as gaseous CO.sub.2 emitted in the flue gas of a power plant. In some other aspects, the industrial source is CO.sub.2 generated during the production of ethanol in an ethanol plant, such as gaseous CO.sub.2 released during the fermentation process. In some other aspects, the industrial source is CO.sub.2 generated during the fermentation of grains during a fermentation process, such as in a brewery. In still other aspects, the industrial source is CO.sub.2 generated during cement production.

[0081] In some other aspects, the source of captured CO.sub.2 **110** for system **100** can be CO.sub.2 captured from the atmosphere.

[0082] Referring now to FIGS. **2** and **3**, a system **100** for the utilization of the source of captured CO.sub.2 **110** as a fuel source **145** for the production of energy is illustrated, wherein the industrial source of CO.sub.2 emissions is a processing plant **101**. In some particular aspects, processing plant **101** has one or more fermenters for the fermentation of a biomass feedstock **105**, particularly plant-based biomass or grains, such as in an ethanol plant or a brewery. Biomass feedstock **105** can be a renewable, non-fossil derived source for production plant **101**. The primary feedstocks of processing plant **101** are biomass feedstock **105**, water, electricity and natural gas.

[0083] During the fermentation of the biomass feedstock **105** in a fermenter of the processing plant **101** gaseous CO.sub.2 is generated. A processing plant **101**, such as an ethanol plant and/or brewery, can also produce gaseous CO.sub.2 during combustion of fuels, such as hydrocarbon

fossil fuels and/or renewable biofuels, for energy production and/or processing purposes.

Accordingly, processing plant **101** can have one or more feedstock streams of captured CO.sub.2 **113**. As shown in FIG. 2, processing plant **101** can have a first feedstock stream of captured CO.sub.2 **113** from a fermenter and a second feedstock stream of captured CO.sub.2 **113** from combustion of fuels. In some preferred aspects, processing plant **101** utilizes a first feedstock stream of captured CO.sub.2 **113** from a fermenter and a second feedstock stream of captured CO.sub.2 **113** from combustion of fuels.

[0084] In some aspects, feedstock stream of captured CO.sub.2 **113** is stored prior to being transformed into fluid CO.sub.2 **123** having a higher energy state. In some other aspects, feedstock stream of captured CO.sub.2 **113** is continually transformed into fluid CO.sub.2 **123** having a higher energy state. In some other aspects, feedstock stream of captured CO.sub.2 **113** is intermittently transformed into fluid CO.sub.2 **123** having a higher energy state.

[0085] In some preferred aspects, processing plant **101** is an ethanol plant for the fermentation of a biomass feedstock **105**. The biomass feedstock **105** can comprise corn, sugar cane, bagasse, miscanthus, sugar beet, sorghum, grain, switchgrass, barley, hemp, kenaf, potatoes, sweet potatoes, cassava, sunflower, fruit, molasses, stover, wheat, straw, cotton, algae, and mixtures or combinations thereof. In some preferred aspects, the biomass feedstock is corn, more preferably kernels of corn.

[0086] One of ordinary skill will appreciate that prior to being harvested, the biomass feedstock **105** utilizes atmospheric CO.sub.2 **103** from air during photosynthesis to grow the biomass that is ultimately harvested as the biomass feedstock **105**. By air, it is meant atmospheric air, which comprises about 78.084% nitrogen, 20.946% oxygen, 0.9340% argon, 0.0416% CO.sub.2, and about 0.1% other components including neon, helium, methane, krypton and water vapor. As such, the biomass feedstock **105** extracts or captures atmospheric CO.sub.2 **103** from atmospheric air and binds the carbon into the biomass during the growing process prior to being utilized as a biomass feedstock **105** in a fermentation process. As such, the biomass feedstock **105** provides a net removal of CO.sub.2 from the atmosphere.

[0087] According to certain embodiments, the system **100** comprises a means for capturing a source of CO.sub.2 emissions from the processing plant **101**, which may include the gaseous CO.sub.2 generated during the fermentation process, combustion of fuels, or a combination thereof, to provide the feedstock stream of captured CO.sub.2 emissions **113**.

[0088] In some preferred aspects, the gaseous CO.sub.2 generated during the fermentation process can be fed directly to the means for transforming the feedstock stream **113** of captured CO.sub.2 into the CO.sub.2 fluid **123** having the higher energy state, such as compressor **120**, as the gaseous CO.sub.2 generated during the fermentation process does not have any of NO_x or SO_x pollutants. In some preferred aspects, the gaseous CO.sub.2 generated during the fermentation process can undergo a purification process to remove impurities, such as a dedusting, removal of oxygen, removal of water, or a combination thereof, prior to being fed to the means for transforming the feedstock stream **113** of captured CO.sub.2 into the CO.sub.2 fluid **123** having the higher energy state, such as compressor **120**.

[0089] In some preferred aspects, the gaseous CO.sub.2 generated from combustion of fuels can undergo a purification process to remove pollutants or impurities, such as desulphurization, denitrogenation, dedusting, or a combination thereof, prior to being fed to the means for transforming the feedstock stream **113** of captured CO.sub.2 into the CO.sub.2 fluid **123** having the higher energy state, such as compressor **120**. In some preferred aspects, the purification process comprises chemical absorption, physical absorption, membrane separation, chemical loop, or a combination thereof, to provide the feedstock stream **113** of captured CO.sub.2 derived from combustion of fuels, such as from a flue gas.

[0090] As such, the one or more feedstock streams of captured CO.sub.2 emissions **113** can be provided to a means for transforming the captured CO.sub.2 emissions into a physical form having

a higher energy state as discussed above. In some preferred aspects, the higher energy state comprises CO.sub.2 fluid **123**.

[0091] During the transformation of the captured CO.sub.2 emissions into a higher energy state, such as the CO.sub.2 fluid **123**, which can be stored as a fuel source **145**, the process can give off considerable heat. In some aspects, as shown in FIGS. **2** and **3**, system **100** can have one or more heat exchange stages **125**, which can be circulated back to the processing plant **101** as a heat source. A first heat exchange stage **125a** will provide a higher temperature than a second heat exchange stage **125b**. In some preferred aspects, compressor **120** comprises a multi-stage compression process with cooling between stages to limit the power needed to drive compressors, allowing for the reduction of size and helps with heat exchanging.

[0092] As previously discussed, CO.sub.2 fluid **123** having the higher energy state can be fuel source **145** in a storage reservoir **140** until fed into a consumptive means **150** that is capable of using the stored potential energy of fuel source **145** to produce an output energy **160** and a discharged CO.sub.2 **170**.

[0093] In some aspects, discharged CO.sub.2 **170** or a portion thereof is released into the atmosphere, which can then be reused by biomass as an atmospheric source of CO.sub.2 **103**.

[0094] In preferred aspects, the CO.sub.2 is consumptive when used as fuel source **145**, such that there is an open loop process providing system **100** as an open system.

[0095] Referring now to FIG. **3**, a system **100** for the utilization of the source of captured CO.sub.2 **110** as a fuel source **145** for the production of energy is illustrated, wherein system **100** has one or more separators **115**. Separator **115** can be located prior to the means for transforming the feedstock of the captured CO.sub.2 **113** into a physical form having a higher energy state, such as compressor **120**. Separator **115** can be located after the means for transforming the feedstock of the captured CO.sub.2 **113** into a physical form having a higher energy state, such as compressor **120**.

Alternatively, at least a first separator **115** can be located prior to and at least a second separator **115** can be located after, the means for transforming the feedstock of capture CO.sub.2 **113** into a physical form having a higher energy state, such as compressor **120**. Separator **115** located prior to the transformation process can remove any undesired components, such as water vapor or the like, to provide a desired CO.sub.2 % in the feedstock of capture CO.sub.2 **113**. Similarly, separator **115** located prior to the storage reservoir **140** can remove any undesired components, such as water vapor or the like, from CO.sub.2 fluid **123**, such that fuel source **145** has a desired CO.sub.2 %. Components that may be corrosive to the system or otherwise affect desirable temperature and/or pressure values of the fuel source **145**, may also be removed by one or more separators **115**.

[0096] Various embodiments of systems, devices, and methods have been described herein. These embodiments are given only by way of example and are not intended to limit the scope of the claimed inventions. It should be appreciated, moreover, that the various features of the embodiments that have been described may be combined in various ways to produce numerous additional embodiments. Moreover, while various materials, dimensions, shapes, configurations and locations, etc. have been described for use with disclosed embodiments, others besides those disclosed may be utilized without exceeding the scope of the claimed inventions.

[0097] Persons of ordinary skill in the relevant arts will recognize that the subject matter hereof may comprise fewer features than illustrated in any individual embodiment described above. The embodiments described herein are not meant to be an exhaustive presentation of the ways in which the various features of the subject matter hereof may be combined. Accordingly, the embodiments are not mutually exclusive combinations of features; rather, the various embodiments can comprise a combination of different individual features selected from different individual embodiments, as understood by persons of ordinary skill in the art. Moreover, elements described with respect to one embodiment can be implemented in other embodiments even when not described in such embodiments unless otherwise noted.

[0098] Although a dependent claim may refer in the claims to a specific combination with one or

more other claims, other embodiments can also include a combination of the dependent claim with the subject matter of each other dependent claim or a combination of one or more features with other dependent or independent claims. Such combinations are proposed herein unless it is stated that a specific combination is not intended.

[0099] Any incorporation by reference of documents above is limited such that no subject matter is incorporated that is contrary to the explicit disclosure herein. Any incorporation by reference of documents above is further limited such that no claims included in the documents are incorporated by reference herein. Any incorporation by reference of documents above is yet further limited such that any definitions provided in the documents are not incorporated by reference herein unless expressly included herein.

[0100] For purposes of interpreting the claims, it is expressly intended that the provisions of 35U.S.C. § 112(f) are not to be invoked unless the specific terms “means for” or “step for” are recited in a claim.

Claims

1-2. (canceled)

3. A system for the capture and utilization of carbon dioxide emissions as a fuel source for the production of energy, the system comprising: a feedstock stream comprising captured carbon dioxide emissions; a first device that is capable of transforming the captured carbon dioxide emissions of the feedstock stream into a carbon dioxide fluid, wherein the carbon dioxide fluid has a higher energy state than the captured carbon dioxide emissions; a vessel for receiving and storing the carbon dioxide fluid as a fuel source; a second device capable of using the fuel source to produce an output energy and a discharged carbon dioxide, the discharged carbon dioxide having a different physical form than the carbon dioxide fuel; and an optional means for recapturing the discharged carbon dioxide.

4. A method of capturing and utilizing captured carbon dioxide emissions as a fuel source for the production of energy, the method comprising: capturing carbon dioxide emissions from a source to provide a feedstock stream comprising captured carbon dioxide emissions; feeding the feedstock stream to a compressor having an input energy source; transforming the captured carbon dioxide emissions of the feedstock stream into a carbon dioxide fluid having stored potential energy, the carbon dioxide fluid having a different physical form than the carbon dioxide gas; storing the carbon dioxide fluid as a fuel source in a usable state to maintain the stored potential energy of the carbon dioxide fuel source; producing an output energy and a discharged carbon dioxide from the stored potential energy of the carbon dioxide fuel source, the discharged carbon dioxide having a different physical form than the carbon dioxide fuel source; and optionally recapturing the discharged carbon dioxide.

5. The system of claim 3, wherein a source of at least a portion of the feedstock stream comprising captured carbon dioxide emissions provided from a fermenter from an ethanol plant or a brewery.

6. The system of claim 3, wherein a source of at least a portion of the feedstock stream comprising captured carbon dioxide emissions is provided as a biogenic waste stream from a biomass feedstock during fermentation of the biomass feedstock at an ethanol plant or a brewery.

7. The system of claim 6, wherein the biomass feedstock is selected from the group consisting of corn, sugar cane, bagasse, miscanthus, sugar beet, sorghum, grain, switchgrass, barley, hemp, kenaf, potatoes, sweet potatoes, cassava, sunflower, fruit, molasses, stover, wheat, straw, cotton and algae.

8. The system of claim 6, wherein the biomass feedstock comprises corn kernels.

9. The system of claim 3, wherein a source of at least a portion of the feedstock stream comprising captured carbon dioxide emissions is provided combustion of one or more hydrocarbon fossil fuels and/or renewable fuels.

10. The system of claim 3, wherein the feedstock stream is transformed into the carbon dioxide fluid via a compressor.
 11. The system of claim 3, wherein the first device utilizes an energy input for transforming the captured carbon dioxide emissions of the feedstock stream into the carbon dioxide fluid, wherein at least a portion of the energy input is a renewable energy input from wind power, solar power, the output energy, or a combination thereof.
 12. The system of claim 3, wherein at least a portion of the carbon dioxide fluid comprises liquid carbon dioxide.
 13. The system of claim 3, wherein a turbine, a turboexpander, or a combination thereof, is capable of using the fuel source to produce an output energy and a discharged carbon dioxide.
 14. The system of claim 3, wherein at least a portion of the discharged carbon dioxide comprises gaseous carbon dioxide, solid carbon dioxide, or a combination thereof.
 15. The system of claim 3, wherein the output energy is in the form of electricity.
 16. The system of claim 3, wherein the output energy is in the form of electricity capable of being fed into an electrical power grid.
 17. The system of claim 3, wherein the output energy is produced by utilizing the higher second energy state of the carbon dioxide fuel source in a turbine to produce electricity, and the discharged carbon dioxide having a third energy state that is lower than the second energy state of the carbon dioxide fuel source.
 18. The system of claim 3, wherein a source of heat is generated during the process of transforming the captured carbon dioxide emissions of the feedstock stream into the carbon dioxide fluid.
 19. The system of claim 3, further comprising at least one heat exchanger, wherein the at least one heat exchanger is capable of utilizing the source of heat generated during the process of transforming the captured carbon dioxide emissions of the feedstock stream into the carbon dioxide fluid.
 20. The system of claim 3, further comprising at least one heat exchanger, wherein the at least one heat exchanger is capable of utilizing the source of heat generated during the process of transforming the captured carbon dioxide emissions of the feedstock stream into the carbon dioxide fluid, wherein the least one heat exchanger utilizes the source of heat in a different location within an ethanol plant or a brewery.
 21. The system of claim 3, wherein the fuel source is stored below its critical temperature and above its critical pressure,.
 22. The system of claim 3, wherein the fuel source is stored above its critical temperature and above its critical pressure, such that at least a portion of fuel source is stored in the form of a supercritical liquid, the fuel source preferably stored above a temperature of 31° C. and a pressure between about 73.8bar and about 125 bar.
 23. The system of claim 3, wherein the discharged carbon dioxide is recaptured as a commercial carbon dioxide source, wherein the commercial carbon dioxide source chosen from the group consisting of industrial grade carbon dioxide having a purity of at least 99.5%, medical grade carbon dioxide having a purity of at least 99.5%, bone dry grade carbon dioxide having a purity of at least 99.8%, food grade carbon dioxide having a purity of at least 99.9%, beverage grade carbon dioxide having a purity of at least 99.9%, anaerobic grade carbon dioxide having a purity of at least 99.95%, or research grade carbon dioxide having a purity of at least 99.999%.
 24. The system of claim 3, wherein at least a portion of the discharged carbon dioxide is recaptured and reused as a portion of the feedstock stream.
 25. The system of claim 3, wherein the captured carbon dioxide emissions of the feedstock stream is transformed into the carbon dioxide fuel having stored potential energy during a period of low energy demand and the fuel source is used for energy production during a period of high energy demand.
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